

Selection Method of AI - LLM for processing Big Data in the Development of Cities Strategic Environmental Documents

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Abstract.

This article presents MALLM (Method for Assessing the Usefulness of Large Language Models), a framework designed to evaluate the applicability of LLMs in developing strategic documents, with a focus on digital transformation policies for urban environmental departments. The growing complexity of urban development, driven by increasing populations and the need to meet Sustainable Development Goals, necessitates effective tools for policy design and analysis. The proposed method supports the selection of appropriate LLMs for use in Retrieval Augmented Generation (RAG) systems.

The study is structured in five parts. First, it identifies a research gap in the use of MALLM for policy document analysis. Second, it introduces the methodology, including indicators, metrics, and evaluation criteria. The approach is demonstrated using environmental data from Gdańsk and Berlin. Selected excerpts from a digital transformation policy are analyzed using four LLMs: LLaMA 3 70B Instruct, Nemotron-4 340B Instruct, Mistral Nemo 12B Instruct, and Phi-3 Medium 128K Instruct. Model performance is evaluated using ROUGE and BLEU metrics, incorporating parameters such as token count and temperature.

The results demonstrate the effectiveness of MALLM in selecting suitable LLMs for policy analysis. The paper concludes with recommendations for applying LLMs in urban policy development and highlights challenges related to Big Data in environmental governance.

Keywords: Large Language Models, MALLM, Digital Transformation Policy, Urban Development, Environmental Governance, RAG Systems, ROUGE, BLEU, Big Data, Policy Analysis

1 Introduction

In the era of generative artificial intelligence, the role of strategic documents is changing. On the one hand, strategic documents will not have as significant a role in the development of society as before due to the dynamics of the changing environment. On the other hand, most strategic documents will be able to be prepared using generative artificial intelligence models [1, 2].

Strategic documents are prepared for various types of organizations, both commercial and public. An important example of public organizations are cities. According to the UN, today more than half of the world's population lives in cities, and by 2050 this indicator will increase to 68% [19]. Goal 11 of the UN Sustainable Development Goals is also entirely devoted to cities [20].

In order to manage such a complex organization as cities, strategies are created for them, both general for the entire city and area-specific for selected areas of their activity. Today, one of the three areas for which strategies are most often created is the environment. An area covering a huge amount and types of data (from air pollution to the carbon footprint of buildings), which are constantly changing. Hence the need to create the digital transformation policies for departments dealing with the environment in cities.

Digital transformation policies show the rules and principles of organizational change in the era of digital technology implementation. The role of technologies and their impact on changes in the organization's business processes, as well as the assessment of the organization's state of maturity, significantly affect the scope and importance of such documents for organizations. Digital transformation policies have a special role for cities, where the city as a public institution, on the one hand, seeks to implement or update these policies on an ongoing basis and, on the other hand, prepare or evaluate any city development documents so that these documents are consistent with the policies [5, 6]. This status of digital transformation policies of cities requires consideration of the place and role of LLM both in the analysis of digital transformation policies and in the way city documents are prepared for the development of these policies.

An analysis of the literature [4] shows that such an approach (of using LLM) to measure the digital transformation process of companies has been proposed. Annual reports of listed companies from 2005 to 2022 were analyzed, and a set of indicators of the digital transformation process was constructed. It was also assumed that current text analysis methods for accurately measuring the progress of corporate transformation pose significant limitations. This was particularly evident in financial reports, where companies often mentioned advanced technologies which did not translate into their application in production processes. To address these shortcomings, a data processing strategy based on large language models (LLM) was adopted.

In turn, the paper [1] presents the effects of digital transformation in education and the use of LLM for this purpose. It was assumed that digital transformation in education is the use of various technological tools and strategies to improve learning outcomes in all areas of the educational system. For digital transformation defined in this way, examples of the use of ChatGPT in education were presented showing how ChatGPT

can provide insights into the learning process which allows modifying the teaching process and improving student performance. The role of artificial intelligence in assessing the work of creating intelligent learning systems was also pointed out. Critical documents were analyzed in which individual suggestions published in newspapers, author's blogs in journals were evaluated in which the integration of ChatGPT into the education sector in emerging economies was shown. The potential benefits of implementing Chat GPT, as well as the drawbacks and risks of using it, are pointed out. In this case, the process of digital transformation is supported by the analysis of the evaluation documents of the teaching process.

The paper [2] presents a model for an enterprise's digital transformation to Industry 4.0. It points out the significant role of the change management process in the course of this process, and examines the extent to which existing digital transformation models and roadmaps for the transformation to Industry 4.0 incorporate lessons learned from change management. A systematic review of the literature in this area from 2017-2021 was presented where various digital transformation models and their support processes were discussed. None of the digital transformation models of the 16 presented in the paper were supported by LMM systems. The significant role of Lean Thinking philosophy was pointed out, the digital Lean transformation framework was shown, and business process management was discussed as a key aspect of digital transformation.

The paper [5] identified research areas of digital transformation and also existing strategies, practices and trends. About 5,000 articles from 2013 to 2022 were analyzed, The probabilistic Dirichlet allocation algorithm and probabilistic topic modeling technique were applied. 34 topics were identified and classified. The dynamics of change of each area of digital transformation was also indicated. This is an example of another article that describes digital transformation processes, identifies areas of transformation, but does not address the topic of manufacturing support. This article shows what potential there is in LLM for analyzing digital transformation processes.

The three representative articles presented above on the state of development of digital transformation policies show that the use of LLM for the first two articles has a *raison d'être* in support of a specific area of digital transformation. The example of the last article aggregating papers in this area shows that the use of LLM to support the construction of these policies was basically not analyzed between 2013 and 2022. This is an example of how a significant area of use of LLM models can be to support the analysis of both the development of digital transformation policies and other documents built for these policies. A review of nearly 5,000 existing articles in [5] indicates that it is becoming necessary to analyze the possibility of using LLM for policy documents. Therefore, the following will provide examples of publications that indicate the potential of large language models to support policy documents. In [3] shows the importance of language models in business and social transformation based on artificial intelligence. The state of language models and their capabilities for text mining are presented.

Different types of language models are presented and the importance of their selection for document analysis is indicated. The article indicates both the importance of neural representation of words using neural models, traditional neural models, deep learning

transfer mechanisms and transfer models, but also the latter Transformers-type models so important in terms of learning mechanisms and use for strategic documents. Also pointed out are the application areas of LLM which is natural language understanding, text extraction in information systems research text mining, enhancement of existing applications for communication with language models, emotion detection based on delivered text and text classification so important for topic modeling processes.

With reference to articles on digital transformation [1, 2, 5], this one presented shows the potential of linguistic models for the analysis of texts so important for the analysis of digital transformation policies. Examples of applications of LLM models but also the classification of these models indicate the need for preliminary analysis of these models for later use.

In [9], the environmental impact of LM use is presented. The first aspect is the increased energy and thus resource consumption and carbon dioxide production that results from the training and subsequent use of LM models. Hence, the authors point out that this is another technology whose carbon footprint should be analyzed. Another of the aspects is the issue of the level of expertise generated by generative artificial intelligence taking into account the level of information, training in LM, and the issue of the form of response in which the results are proposed.

Therefore, it would be appropriate to consult the received results with expert opinions and assess their reliability. More worrisome in this case is the possibility of unintentional or intentional introduction of systematic errors concerning both the training, the processing algorithm and the form of the obtained results. The authors note that with many positive aspects regarding the use of LM models, the use of LLM involves environmental risks. Therefore, it is necessary to start preliminary discussions about the risks of using LLMs at this stage, and therefore at this point it would be appropriate to consider what kind of business model should be adopted to take into account both the opportunities and risks of using these models.

In [7], the potential benefits and challenges of LLM educational applications are presented from the perspective of students and teachers. The current state of LLMs and their applications creating educational content, enhancing student engagement and interaction, and personalizing learning experiences are presented. A clear strategy for implementing educational systems with a strong focus on critical thinking and fact-checking strategies received from LLM is required for such approaches. These models use the Transformer architecture and have been pre-trained on massive data sets.

The collection of representative publications presented above, which summarize the state of knowledge on the creation of digital transformation policies and the possibilities of their support [1, 2,5] as well as the possibilities of using LLM [3, 9, 7], shows how diverse these areas are in research on the construction of digital transformation policies their analysis and monitoring the status of their use. This is a kind of research gap the filling of which will indicate how important it is to select appropriate LLM models for private construction. Therefore, the next chapter will present a method for selecting language models that can be used for the development of digital transformation policies and subsequent use for the creation of city documents.

Article [21], presented at the EnviroInfo 2024 conference, serves as a reference point for the considerations raised in the proposed second article, providing theoretical foundations and preliminary experimental results regarding the use of large language models (LLMs) in the analysis of environmental documents. Specifically, it presents an approach based on multilingual modeling in the context of ESG reporting and demonstrates the potential of LLMs in assessing document compliance with electronic environmental metrics in various countries. A significant contribution of this article is the identification of the problem of selecting an appropriate language model and the importance of fine-tuning it in the context of the specific data and regulations being analyzed.

Against this backdrop, the current article expands on and develops a methodological approach that systematizes and formalizes the issue of LLM model selection. In response to the research gap identified in article [21] regarding the lack of a structured model selection method, the authors propose the MALLM (Method for the Assessment of the Usability of Large Language Models). This method expands on the previous approach by introducing measurable evaluation criteria (including the Rouge and Bleu metrics) and incorporating model performance parameters such as token count and temperature, enabling multidimensional analysis of their performance. Furthermore, the current proposal goes beyond the scope of applications presented in article [21], shifting the focus from ESG reporting to processing Big Data in strategic urban development documents. This represents a shift from document compliance analysis to supporting decision-making and planning processes in public administration. In this context, LLMs no longer serve solely as analytical tools but are becoming an integral component of Retrieval Augmented Generation (RAG) systems that support the creation and modification of public policies. The current proposal also introduces a comparative component by analyzing different LLM models in a controlled experimental environment, a direct extension of the conclusions from article [21] regarding the need for their selection. This shifts the approach from an exploratory to a systematic and replicable one.

2 Materials and methods

This section of the article presents the MALLM method of selecting LLM models to assess the suitability of LLMs for developing digital transformation policies for cities. The experiments conducted and described below showed that this can be a preliminary step to a still-developed method for creating expert and formal mechanisms for assessing the suitability of LLMs. This means that the method presented in the article requires further research. To make the description more reader-friendly this chapter is divided into three main parts.

The first presents the assumptions for the developed MALLM method. The resources that were used for the implementation and development of the developed method were shown. These include model repositories, but also metrics used in the evaluation of MALLM as well as digital transformation policy documents. These three elements

(repositories, metrics and policies) are described in this part of the work, in order to then present the post-processing steps developing the method.

In the second part, the method was presented when evaluating the usefulness of LLM. The stages and principles of the development of the method for assessing the suitability of LLM for analyzing digital transformation policies were presented. In turn, part three summarizes the status of these stages and principles of the method under development.

2.1 Assumptions for the development of the MALLM method for Big Data resources

More and more European municipalities, such as Berlin and Gdansk are operating open data portals. On a voluntary basis, they post official data on many - in the best case, all - administrative areas on these portals. Citizens, business, science and other administrations are able not only to find out about the available data.

The open data portal of city Berlin <https://daten.berlin.de/> obtains Big Data resources on a large variety of categories as they are named in "Smart City Charta" <https://www.bbsr.bund.de/BBSR/DE/veroeffentlichungen/sonderveroeffentlichungen/2017/smart-city-charta-de-eng-dl.pdf> :

- "Smart Energy & Environment" with focus on renewable energies, energy efficiency and environmental protection
- "Smart Living" with focus on improving living comfort through networked household appliances and intelligent building technologies
- Smart Infrastructure with focus on development of modern, efficient and intelligent infrastructures
- Smart Health Care enabling access to digital health services and e-health solutions
- Smart Governance is focused on digitization and improvement of processes between citizens and administration
- Smart Transport & Mobility for efficient and resource-saving mobility solutions, including sharing services and autonomous vehicles
- Smart People for promoting digital education and providing digital services for all citizens
- Smart Economy is using the city's innovative economic power to promote start-ups and technological developments.

This work is using exemplarily the data set for air quality measurement of fine dust PM10, fine dust PM2.5, nitrogen dioxide NO₂, nitric oxide NO, nitrogen oxides NO_x and ozone O₃ in µg/m³ with hourly frequency from 4th of May until 4th of June 2025 from open data portal https://odis-berlin.de/aktuelles/2023-12-21-luftguete/?utm_source=chatgpt.com#luftguetemesdaten-als-offene-datenschnittstelle

Gdansk

The city of Gdansk has had a defined open data policy for over 8 years. It has a dedicated platform, Open Gdansk (<https://www.gdansk.pl/otwarte-dane-w-gdansk>), which contains over 50 data sets divided into 11 categories. One of them is the "environment" category, which presents data on:

- Waste generated in the city
- Land in the city
- Quality of drinking water and sewage
- Air quality
- List of natural monuments

The air quality collection contains data on pollutant emissions, average annual sulfur dioxide concentration, average annual nitrogen dioxide concentration. Historical data is available, but also data generated live from air quality measurement stations located in different parts of the city.

The LLM evaluation process required a detailed analysis of existing resources of pre-trained LLMs. Therefore, at the outset, eight repositories containing pre-trained models were extracted. The resources used were: Hugging Face, TensorFlow Hub, PyTorch Hub, Model Zoo by AllenNLP, Spacy Models, Google Cloud AI, AWS Marketplace, Microsoft Azure Cognitive Services.

The selection of the repository was guided by a preliminary assessment of the applicability of LLM for policy document analysis. Therefore, from the LLM functionalities extracted according to the Hugging Face repository: text analysis, text classification, sentiment analysis, machine translation, information extraction, text summarization, question answering, sentiment analysis and public opinion, for text generation, those were selected that, can be used for the development of digital transformation policies: models for generating arbitrary texts

Models for developing dialogues and chatbots were also considered due to the possibility of constant interaction with users in the form of questions and answers. Models for answering questions were also considered due to their focus on providing answers to the questions asked about the stages of digital transformation analysis. On this basis, four models were qualified for further study: llama3-70b-instruct, nemotron-4-340b-instruct, mistral-nemo-12b-instruct, phi-3-medium-128k-instruct.

The next step in building the assumptions for the construction of the MALLM was to analyze the metrics used in the LLM evaluation. The following metrics were considered:

- BLEU (Bilingual Evaluation Understudy Score).
- ROUGE (Recall-Oriented Understudy for Gisting Evaluation).
- METEOR to analyze different linguistic forms.
- TER (Translation Edit Rate), which is used to measure the number of edits needed to convert the generated text into reference text.

Two metrics, BLEU and ROUGE, were selected for further study based on expert analysis. The BLEU metric concerns n-gram matching (generated text and reference text) and is used in machine translation and text generation. ROUGE, on the other hand, analyzes longer common (generated text and reference text) subsequences, which is particularly useful for evaluating summaries and text generation.

The digital transformation policy for the City of Warsaw, built by the authors of the article, was used for the research. After the digital transformation policy was handed over, public consultations were held for 8 months with citizens, employees of offices

and institutions, during which the directions, principles and values existing in the policy were confirmed or supplemented. On December 29, 2020. The Mayor of Warsaw issued Ordinance No. 1494/2020 on the implementation of the Digital Transformation Policy for Warsaw. To assess the suitability of the LLM for analyzing the digital transformation policy, the following questions were posed to the LLM:

- How you build a city's digital transformation using a strategy document - a city's digital transformation policy focused on citizen engagement.
- Can digital transformation be treated as an iterative process and implemented in organizations and IT teams. The perspective (project, process, technology and organizational) of digital transformation is used in transformation processes to achieve the goals of these organizations and teams.
- How to build and develop a digital transformation policy together with city residents and the city hall IT organization?
- What processes or process models and information technologies that support organizational development can be used to build digital transformation policies?
- Does the use of process models based on reverse engineering and information technology to support organizational development based on Enterprise Architecture create conditions for building a process model of the city's digital transformation policy?

For the questions posed, texts were generated using LLM and compared with reference texts included in the digital transformation policy developed for Warsaw. The reference answer included in the digital transformation policy for question 1 is shown below:

"Building a city's digital transformation using a digital transformation policy document focused on citizen engagement begins with identifying the directions, values and principles of transformation. The first stage uses the city's own best practices in developing digital transformation policies, and then identifies and applies case studies of digital policy implementation in other cities. The case studies make it possible to aggregate cities' digital transformation processes and confront them with the situation in the city under study. It is also important to involve citizens in the method, which helps accelerate innovation and create new directions for city transformation processes. The process is iterative and can be monitored by city residents, such as NGOs, through the use of enterprise architecture and reverse engineering, allowing for better control of the strategy document and its development."

2.2. MALLM method for assessing the suitability of LLM for developing digital transformation policies

For the assumptions thus made, the presentation of the MALLM method proceeded. An approach was used in which eight stages of method development were defined. Existing MALLMs [18] were given a preliminary evaluation based on descriptions, literature analysis and usage. Therefore, the developed MALLM method is presented below in Figure 1

The process involves eight sequential steps with feedback. The method means that the selection of LLMs for strategic process analysis should be a continuous process in which, models that do not meet the requirements of the Blue and Rouge indicators of LLM models or new emerging LLMs should be subject to a process of continuous evaluation of their suitability for analyzing digital transformation policies.

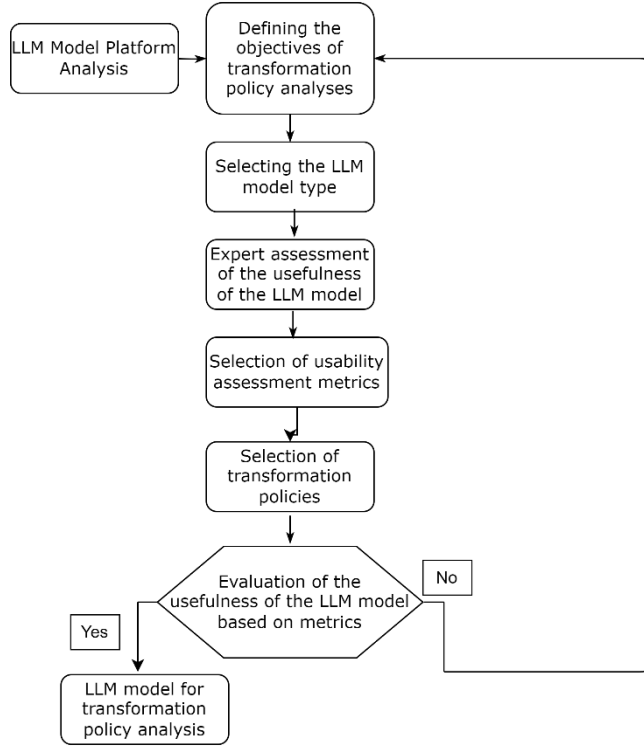


Fig.1 Developed MALLM method of selecting LLM models to assess their suitability for analysis of digital transformation policies

Stage one includes an analysis of the platforms and repositories that host LLM models. We take into account both repositories and libraries that provide access to LLM so that all available models can be considered in the evaluation process. In a situation in which we have a full understanding of LLM models we proceed in stage two to the definition of the goals that face the analysis of digital transformation policies. In this case, it is necessary to consider the expectations of such analyses. Are we interested in the correctness of the preparation of the policy? or in its development? or perhaps in the use of the policy for the needs of the city.

This is a key stage in the development of the method in which clearly defined objectives will allow us to isolate the right LLMs on the one hand and to conduct the evaluation process correctly on the other. Having available platforms with models and defined goals of digital transformation policies, we proceed to the selection of the

type of LLM. It should be noted at this point that we are not interested in specific LLMs in this case but in the types of LLMs, i.e. those that are designed, for example, for text analysis, questioning or others. Details of the selection are presented in the assumptions for building the MELLM

At this stage, it seems necessary to proceed with an expert assessment of suitability based on descriptions of models available in repositories and LLM platforms. In this case, it is important to recognize the potential of the community, which conducts such assessments very often. Maybe not such comprehensive assessments, which are the content of this article, but assessments that provide a basis for subjective selection of a given LLM. At this stage of the research, no formal evaluation methods like multi-criteria analysis were also proposed, assuming that we are at the stage of preliminary construction of the MALLM.

The next stage involves the selection of metrics that will form the basis for evaluating the content generated by the LLM and the content of the proposed policies. In this case, we are interested in either the precision of the responses or the context of the LLM responses. In the next stage, we select the texts that are in the digital transformation policies that will form the basis for evaluating the usefulness of the LLM. At this stage of the research, it seems that an expert selection of texts, preceded by an analysis of these texts and their relevance to digital transformation policies, will be the most complete.

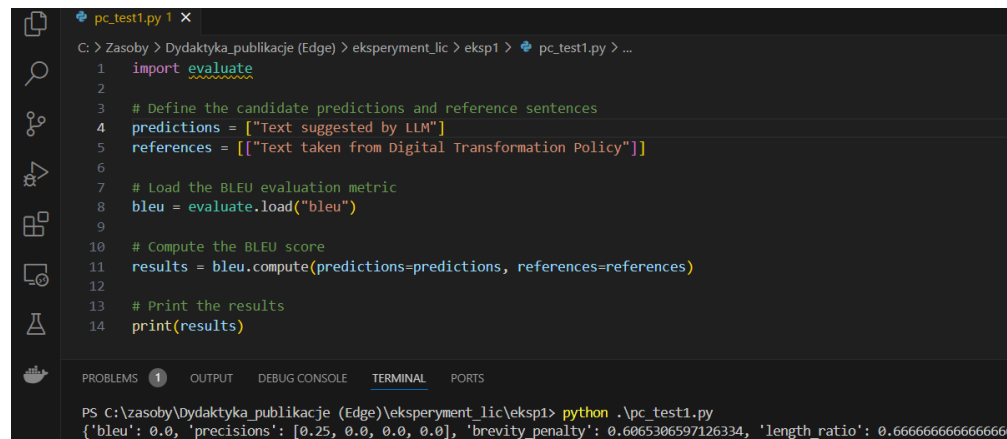
Having selected LLMs, evaluation metrics, and texts to be evaluated in the last two stages of MALLM, we proceed to evaluate relevance using the proposed metrics. It has been assumed, and this will form the basis of the discussion later in the article, to consider whether it would be worthwhile for us to consider developing a single selected relevance metric. The experiment conducted, the results of which will be presented below, will show proposals for such an aggregate index.

3 Results

This section of the article will present the results of evaluating four LLM models using Bleu and Rouge metrics. The chapter is divided into three main parts. In the first part, the selection of models is shown in the second part the application of metrics to carry out on evaluation. This part also proposes an index for evaluating digital transformation documents integrating the components of BLUE and ROUGE metrics. The third part presents the results obtained. To verify the developed approach, the Nvidia site was used with the following LLM groups:

- Models designed for inference.
- Multimodal models that can infer from image and video inputs and perform descriptive language generation.
- Models designed for generation.
- Embedding models to connect chat-based LLM models to proprietary data
- Conversion models
- Group of industry models used in healthcare

A group of models for inference was considered and four models were extracted: llama3-70b-instruct, nemotron-4-340b-instruct, mistral-nemo-12b-instruct, phi-3-medium-128k-instruct. Figures 2 and 3 show how to evaluate using the Python language. Lines 4 and 5 include the reference a digital transformation policy and the predictive text proposed by LMM, respectively.



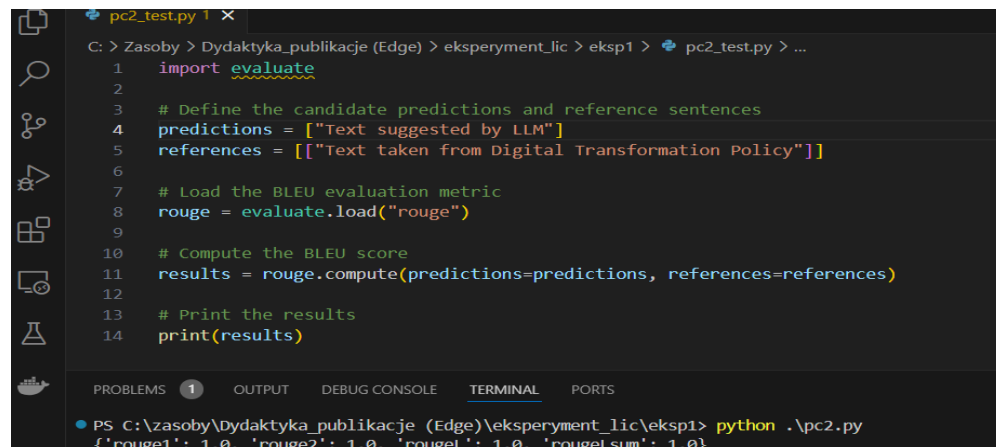
```

pc_test1.py 1 X
C: > Zasoby > Dydaktyka_publikacje (Edge) > eksperyment_lic > eksp1 > pc_test1.py > ...
1 import evaluate
2
3 # Define the candidate predictions and reference sentences
4 predictions = ["Text suggested by LLM"]
5 references = ["Text taken from Digital Transformation Policy"]
6
7 # Load the BLEU evaluation metric
8 bleu = evaluate.load("bleu")
9
10 # Compute the BLEU score
11 results = bleu.compute(predictions=predictions, references=references)
12
13 # Print the results
14 print(results)

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\zasoby\Dydaktyka_publikacje (Edge)\eksperyment_lic\eksp1> python .\pc_test1.py
{'bleu': 0.0, 'precisions': [0.25, 0.0, 0.0, 0.0], 'brevity_penalty': 0.6065306597126334, 'length_ratio': 0.6666666666666666}

```

Fig. 2 Evaluation of LMM suitability using BLEU metrics



```

pc2_test.py 1 X
C: > Zasoby > Dydaktyka_publikacje (Edge) > eksperyment_lic > eksp1 > pc2_test.py > ...
1 import evaluate
2
3 # Define the candidate predictions and reference sentences
4 predictions = ["Text suggested by LLM"]
5 references = ["Text taken from Digital Transformation Policy"]
6
7 # Load the BLEU evaluation metric
8 rouge = evaluate.load("rouge")
9
10 # Compute the BLEU score
11 results = rouge.compute(predictions=predictions, references=references)
12
13 # Print the results
14 print(results)

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\zasoby\Dydaktyka_publikacje (Edge)\eksperyment_lic\eksp1> python .\pc2.py
{'rouge1': 1.0, 'rouge2': 1.0, 'rougeL': 1.0, 'rougeLsum': 1.0}

```

Fig 3. Evaluation of LMM suitability using the ROUGE metric.

The results of the presented evaluations are included in Table 1. They contain indicators of BLEU and ROUGE metrics obtained for different w parameters of LLM - maximum number of tokens and temperature. The token limit affects the length of the proposed text. If the number of tokens is too small, the LLM may not have enough

information to understand the full context, which may lead to less accurate text-prediction responses. Temperature, on the other hand, affects the accuracy of the LLM. Too low a temperature (e.g., 0.2 - 0.4): is suitable for tasks requiring precision, accuracy and consistency. Medium temperature (e.g. 0.5 - 0.7), on the other hand, is a compromise setting that allows predictions to be creative while maintaining relatively high precision. In contrast, high temperature (e.g., 0.8 - 1.0) is used for creative tasks where originality and variety are more important than accuracy. Therefore, Table 1 presents the evaluation results of the models considering different values: the number of tokens and temperature. Table 1 also presents a proposal for an integrated evaluation index. The proposal of the integrated indicator as an average between the results of BLEU and ROUGE-L is a logical compromise because it takes into account different aspects of evaluating predictions. BLEU focuses on precision by comparing word sequences, while ROUGE-L evaluates the matching of word sequences. Combining these two metrics can lead to a more comprehensive evaluation that takes into account both accuracy and fluency of predictions.

Table 1. Results of LLM evaluation using BLEU and ROUGE metrics

Model name	Max Tokens	Temperature	bleu	precisions_1	precisions_2	precisions_3	precisions_4	brevity_penalty	rouge1	rouge2	rougeL	rougeLsum	General Index
llama3-70b-instrukcja	1024	1	0.112	0.331	0.156	0.11	0.137	0.396	0.208	0.0512	0.13	0.13	0.121
	513	1	0.047	0.205	0.051	0.29	0.017	1	0.279	0.086	0.193	0.193	0.12
	129	1	0.001	0.717	0.342	0.157	0.087	0.004	0.417	0.143	0.3	0.3	0.1505
	1024	0.75	0.224	0.663	0.338	0.198	0.14	0.799	0.229	0.067	0.156	0.156	0.19
	513	0.75	0.12	0.711	0.338	0.186	0.133	0.409	0.288	0.806	0.19	0.19	0.155
	129	0.75	0.001	0.717	0.343	0.144	0.087	0.004	0.412	0.143	0.308	0.308	0.1545
	1024	0.5	0.218	0.663	0.333	0.193	0.138	0.778	0.236	0.678	0.156	0.156	0.187
	513	0.5	0.114	0.711	0.337	0.185	0.133	0.409	0.288	0.08	0.19	0.19	0.152
	129	0.5	0.001	0.717	0.343	0.144	0.087	0.005	0.417	0.143	0.308	0.308	0.1545
	1024	0.2	0.218	0.663	0.334	0.193	0.138	0.778	0.236	0.067	0.156	0.156	0.187
	513	0.2	0.114	0.712	0.337	0.186	0.133	0.409	0.286	0.08	0.19	0.19	0.152
	129	0.2	0.001	0.717	0.343	0.144	0.087	0.005	0.417	0.143	0.308	0.308	0.1545
	1024	1	0.131	0.667	0.247	0.097	0.575	0.748	0.27	0.077	0.169	0.169	0.15
	513	1	0.103	0.683	0.254	0.113	0.676	0.537	0.299	0.085	0.182	0.182	0.1425
	129	1	0.002	0.766	0.386	0.228	0.173	0.005	0.029	0.295	0.233	0.233	0.1175
	1024	0.75	0.143	0.643	0.24	0.101	0.056	0.821	0.256	0.073	0.154	0.154	0.1485
nemotron-4-340b-	513	0.75	0.098	0.684	0.248	0.105	0.006	0.541	0.297	0.085	0.181	0.181	0.1395
	129	0.75	0.002	0.765	0.387	0.228	0.173	0.005	0.293	0.096	0.233	0.233	0.1175
	1024	0.5	0.153	0.617	0.226	0.09	0.045	0.97	0.24	0.065	0.146	0.146	0.1495
	513	0.5	0.098	0.684	0.249	0.105	0.06	0.542	0.248	0.085	0.181	0.181	0.1395
	129	0.5	0.002	0.761	0.463	0.271	0.179	0.006	0.408	0.164	0.278	0.278	0.14
	1024	0.2	0.158	0.653	0.279	0.13	0.072	0.776	0.257	0.95	0.172	0.172	0.165
	513	0.2	0.133	0.686	0.299	0.14	0.080	0.609	0.286	0.1	0.186	0.186	0.1595
	129	0.2	0.002	0.761	0.463	0.271	0.179	0.006	0.408	0.164	0.278	0.278	0.14
	1024	1	0.097	0.618	0.205	0.067	0.029	0.771	0.234	0.064	0.138	0.138	0.1175
	513	1	0.041	0.559	0.128	0.039	0.014	0.51	0.285	0.085	0.157	0.157	0.099
	129	1	0	0.587	0.197	0.066	0.022	0.002	0.29	0.089	0.187	0.187	0.0935
	1024	0.75	0.0223	0.265	0.057	0.015	0.006	0.66	0.21	0.063	0.131	0.131	0.07665
	513	0.75	0.023	0.475	0.117	0.032	0.009	0.379	0.252	0.075	0.155	0.155	0.089
mistral-nemo-12b-	129	0.75	0	0.477	0.19	0.063	0.032	0	0.217	0.079	0.158	0.158	0.079
	1024	0.5	0.011	0.454	0.095	0.027	0.008	0.27	0.247	0.079	0.158	0.158	0.0845
	513	0.5	0.037	0.53	0.126	0.039	0.016	0.456	0.234	0.069	0.167	0.167	0.102
	129	0.5	0	0.425	0.081	0.035	0.023	0.001	0.257	0.099	0.168	0.168	0.084
	1024	0.2	0.0211	0.0159	0.033	0.01	0.004	1	0.267	0.067	0.17	0.17	0.09555
	513	0.2	0.012	0.277	0.042	0.012	0.006	0.415	0.178	0.058	0.109	0.109	0.0605
	129	0.2	0	0.419	0.11	0.0416	0.028	0	0.25	0.101	0.18	0.18	0.09
	1024	1	0.276	0.472	0.097	0.014	0.001	0.92	0.237	0.69	0.127	0.127	0.2015
	513	1	0	0.445	0.046	0.003	0	0.291	0.271	0.049	0.145	0.145	0.0725
	129	1	0	0.488	0.108	0	0	0	0.236	0.008	0.142	0.142	0.071
	1024	0.75	0.3	0.26	0.046	0.011	0.005	1	0.272	0.057	0.18	0.18	0.24
	513	0.75	0	0.226	0.035	0	0	1	0.262	0.05	0.17	0.17	0.085
phi-3-medium-	129	0.75	0	0.256	0.012	0	0	0.431	0.211	0.029	0.135	0.135	0.0675
	1024	0.5	0	0.224	0.0218	0	0	0.91	0.24	0.04	0.178	0.178	0.089
	513	0.5	0	0.246	0.027	0	0	0.96	0.27	0.054	0.193	0.193	0.0965
	129	0.5	0	0.262	0.025	0	0	0.41	0.225	0.059	0.167	0.167	0.0885
	1024	0.2	0.03	0.1933	0.04	0.017	0.009	1	0.2	0.064	0.156	0.156	0.093
	513	0.2	0.045	0.188	0.056	0.0256	0.015	1	0.268	0.082	0.193	0.193	0.119
	129	0.2	0.089	0.475	0.184	0.092	0.062	0.59	0.389	0.183	0.279	0.279	0.184

The results presented were ordered by taking into account for each LLM the maximum number of tokens and temperature according to the indicators for BLEU and ROUGE. The analysis of both indicators indicated the need to generate a common indicator that would allow the evaluation of the generated text from two different points of view: similarity and the number of the longest strings of characters that the LLM generates.

An analysis of the attached table shows that as the number of tokens decreases, the BLEU rates decrease. This situation is the case for each of the LLMs analyzed. The analysis of LLM precision: precision one, two, three and four shows the differentiated behavior of LLMs considering the maximum number of tokens and temperature. While in the case of the BLEU indicator, the process of lowering the BLEU occurs as the number of tokens decreases, it is difficult to determine any measure when determining the precision. It is also necessary to consider whether the LLM precision rate and its change affects the values as well as the similarity of LLM behavior considering the decreasing number of tokens

A similar relationship should be observed in the case of the ROUGE indicator where both the value of 1, 2 and L i.e. the longest similar chains is similar but does not show any trend, however. On the other hand, in the case of ROUGE-L, it is clear that the smaller this indicator is, the more difficult it is to use it to assess the suitability of LLM. Thus, taking into account both components does not give the possibility to evaluate the suitability of LLM because the maximum values of BLEU do not correspond to the maximum value of ROUGE considering any number of tokens and temperature.

Therefore, the authors of the paper proposed a solution, whereby the average index of BLEU and ROUGE was adopted, assuming that both indexes refer to the maximum probability assessment of the generated text and the reference text. The adoption of the average index allows to sort out the usefulness of these four models, where the maximum values occur for the model of two LLMs: llama3-70b-instruct and phi-3-medium-128k-instruct. The level of these indicators shows that the predictions generated by the two LLMs are at 20% similarity to the reference texts. Two examples of LLM evaluation interpretations using the BLEU and ROUGE metrics are presented below.

a/ Example of evaluation with BLEU

```
{  'bleu': 0.0886851077452981,  'precisions': [0.47474747474747475,
0.1836734693877551, 0.09278350515463918, 0.0625],  'brevity_penalty':
0.591406000558906,
```

The BLEU value ranges from 0 to 1, where 1 indicates a perfect match. The value of 0.0887 is quite low, suggesting that the model does not generate texts very close to the reference texts.

```
precisions: [0.47474747474747475, 0.1836734693877551, 0.09278350515463918,
0.0625]
```

These are the precisions of the n-grams (i.e., sequences of words of different lengths) in the generated text, compared to the reference texts.

1-gram (single words): 0.4747

2-gram: 0.1837

3-gram: 0.0928

4-gram: 0.0625

These values show what percentage of the n-grams in the generated text are present in the reference texts. As you can see, the precision decreases as the length of the n-grams increases, which is typical.

Brevity Penalty (BP)

BP is close to 1 when the length of the generated text is close to the length of the reference text, and drops when the generated text is much shorter. The reported value of 0.5914 indicates that the generated text is much shorter than the reference text. In summary, the results suggest that the model generates text that is not very close to the reference text, has fairly low n-gram precision, and generates texts that are significantly shorter than the reference texts.

b/ Example of evaluation with ROUGE

```
{ 'rouge1': 0.38983050847457623, 'rouge2': 0.15384615384615383, 'rougeL': 0.2796610169491525, 'rougeLsum': 0.2796610169491525}
```

ROUGE-1 measures the coverage of single words (1-grams) of the generated text relative to the reference text. A score of 0.3898 means that about 39% of the single words from the reference text also appear in the generated text.

ROUGE-2 measures the coverage of two-word sequences (2-grams) of the generated text relative to the reference text. A score of 0.1538 means that about 15% of the two-word sequences from the reference text also appear in the generated text.

ROUGE-L measures the Longest Common Subsequence (LCS) coverage between the generated text and the reference text. A score of 0.2797 means that the longest common subsequences of the generated text overlap with the reference text by about 28%.

ROUGE-Lsum is similar to ROUGE-L, but instead of comparing entire documents, it compares the longest common substrings at the sentence level, which is often used to evaluate multi-sentence summaries. In this case, the result is the same as for ROUGE-L, suggesting that for this data set, the document-level and sentence-level results are identical. In summary, these results suggest that the generated text has some level of coverage with the reference text, but it is not perfect. ROUGE-1 indicates that about 39% of single words agree, ROUGE-2 shows 15% agreement of two-word sequences, and ROUGE-L shows that about 28% of the longest common substrings agree with the reference text.

Based on the selection of the language model, the Llama model (the highest indicator value) was selected to support the development processes of strategic documents. Then, this model was used in the RAG (Retrieval-augmented generation) system - LM Studio (Figure 4).

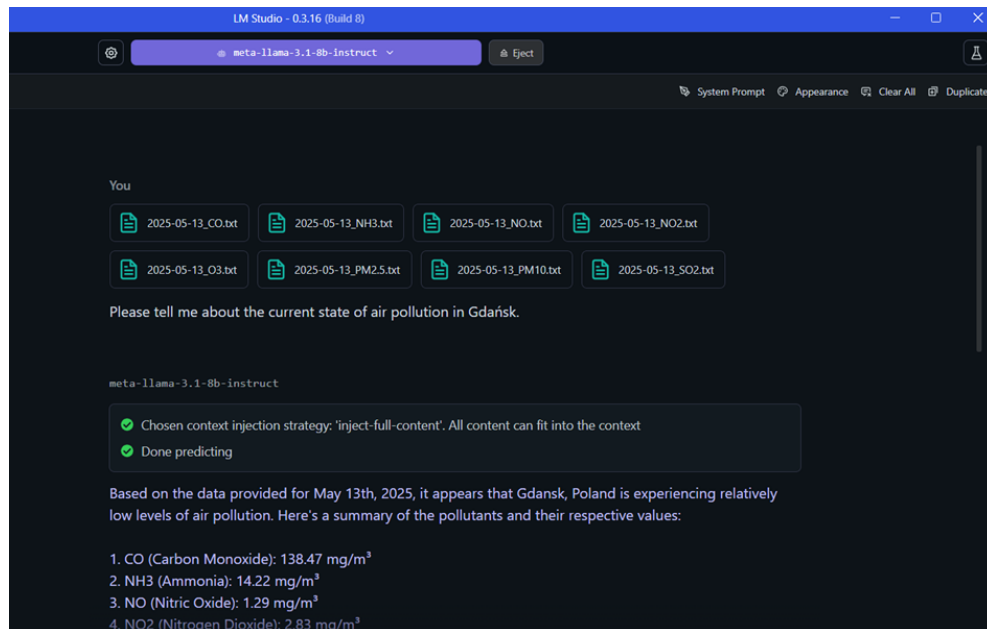


Fig 4. RAG system using the language model selected in the first table to analyze Big Data components - air pollution in Gdańsk

Next, a publisher system was developed that collects data from external air quality measurement stations in Gdańsk. This system implements the tasks of the digital transformation policy - full information on the state of the environment. This data was transferred to the Llama language model in LM Studio. This solution allowed the language model to ask questions about the state of the city's development - air pollution, increase or decrease in the number of measurement stations, development plans). Figure 5 presents the current states of air quality and possible forecasts for the coming days. This solution (integration of data constituting the basis for Big Data and LLM) allows for the integration of strategic documents such as digital transformation policies with Big Data systems in which the data may concern any areas of the city's development.

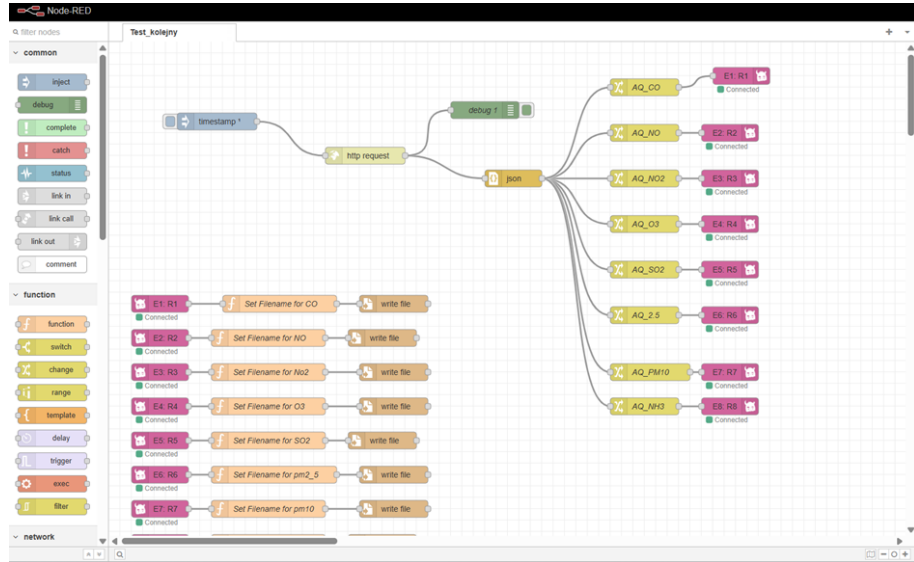


Fig 5. Flow-based (Node-Red) system for downloading Big Data containing detailed data on air pollution in Gdańsk.

4 Discussion

The starting point for the discussion is the analysis of the state of the literature presented in the introduction. It should be noted that in most of the publications presented, policies are implemented using traditional methods, or policies refer to LLM implementation as a stage in the transformation process. This seems to be a standard approach, in which the authors have been using familiar procedures, processes and iterations related to the construction of policies and their use for many years. In turn, the posted publications on the use of LLM models show what potential there is in these models and how it is necessary to look for areas where they can be applied. Therefore, the question arises as to how novel is an approach in which we analyze existing language models, select the ones we find most useful for building digital transformation policies and then evaluate their usefulness from the perspective of metrics and value.

This question formed the basis for the preparation of this article, which presents the method developed by the authors for selecting LLMs to support the analysis of digital transformation policies. The development of the method was based on an expert well type of LLM model and then a formal evaluation of its suitability to support the processes of analyzing digital transformation policies. In the first case, this expert evaluation basically referred to the choice of model type. Those presented in the repositories focus on many different aspects: such as image analysis, hybrid image and text analysis, translation, but also inference.

Therefore, after analyzing existing models, the authors selected a group of models designed for inference to consider to what extent inference processes can support the

analysis of policy documents. The method also included a set of metrics designed to evaluate the usefulness of the models, where the reference text included in the digital transformation policy is compared with the text projected using LLM. This analysis, presented in Table One, shows that both metrics (BLEU and ROUGE) show a differentiated approach to assessing the usefulness of LLM, and choosing one of them would not be sufficient to assess the usefulness of LLM. Therefore, it was chosen by considering an indicator that showed a significant proportion of character strings and word strings that could provide a basis for assessing the similarity of the reference and projected text. On this basis, an integrated index was developed that takes into account the influence of both BLEU and ROUGE models, and on this basis of the four proposed models, two were selected for which the predictive text and reference text were similar at the 20% level.

It remains to be seen to what extent this is a significant value showing the usefulness of LLM models to support the development process and analysis of digital transformation policies. In particular, for the "Environment" area in cities, support in selecting a tool for creating strategic documents is important due to the daily challenges related to the growth of cities and the sustainable development goals that cities must meet. Based on the conducted experiment, it is also necessary to indicate the challenges in the use of Big Data. While cities offer some data in the open data formula (the examples of Berlin and Gdańsk cited), many cities still have documents in the form of data types that are difficult to use automatically (starting with the still used paper data through PDF formats). Hence, it is all the more important to create adequate digital transformation strategies to be able to fully use the potential of the data you have.

This analysis was performed using the Python language and required manual selection of LLMs. It seems that in the next stage of development of the method proposed in the article, the process of both generating the text assessing the similarity of the generated text to the reference text should be automated

5 Conclusions

The authors' research shows that the selection of LLM models for the analysis of strategic texts should be hybrid: formal-expert, in which experts first analyze areas of LLM use and then select LLMs from a given area for which a formal evaluation using metrics is applied.

The MALLM method used in the paper to evaluate the suitability of LLMs based on the BLEU and ROUGE indexes shows that the use of both or one of them does not allow a clear choice of LLM. Therefore, an integrated index that takes into account both the lengths of the strings and the accuracy of the prediction by means of a comparison of the reference and predictive text was proposed in deciding the model selection.

The method developed by the authors should be developed, and therefore the proposed approach uses a feedback process whereby subsequent emerging LLM models should be taken into account in the initial selection of the LLM type and then specialized within the model type to assess its suitability.

It seems that the directions of further research should lead to an increase in metrics from the two that were proposed in the article to the next, but also to an analysis of the usefulness of metrics for assessing the similarity of the reference text and the predicted text. Engrams that appear in both metrics may prove to be limiting in the selection of models.

The research conducted was on digital transformation policies for the area of "environment" in cities and evaluation of the usefulness of LLM. Conclusions can be drawn from this research for analyzing other types of policy documents. However, this requires similar research. After a number of strategic documents, it will then be possible to conclude that the developed method can be used to evaluate any strategic documents and lead to the selection of appropriate models to support their analysis process.

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